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EIGHTH SEMESTER

Roll No. .....

END SEMESTER EXAMINATION

B.E. (IT)

MAY-JUNE 2010

IT-414 ARTIFICIAL INTELLIGENCE

Time: 3 Hour

Max. Marks : 70

Note : Question NO. ***ONE*** is compulsory.
Answer any ***FOUR*** questions from the remaining.
Assume suitable missing data, if any.

1[a] Construct partitioned semantic net representations for the following:
(i) Every batter hit a ball
(ii) All the batters like the pitcher. **5**

[b] What are fuzzy sets? Differentiate between fuzzy set and crisp set. **3**

[c] Is minimax procedure suitable for three to four players? Justify your answer. **3**

[d] Explain A0* algorithm with example. **3**

2[a] Consider the following sentences
→ John likes all kinds of food
→ Apples are food
→ Chicken is food
→ Anything anyone eats and isn't killed by is food
→ Bill eats peanuts and is still alive
→ Sue eats everything Bill eats.
(i) Translate these sentences into formulae in predicate logic. **2**
(ii) Prove that John likes peanuts using resolution. **8**
(iii) Use resolution to answer the question "What food does Sue eat'? **2**

[b] Discuss all the issues in knowledge representation. **2**

3[a] Why must A0* algorithm work properly on graphs containing cycles? Justify your answer. **4**

[b] Consider trying to solve 8 puzzle using hill climbing. Find heuristic function that makes the puzzle work.

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8	5	6
4	7	

Start

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Goal

10

4[a] Differentiate between forward and backward reasoning with example? **4**

[b] Explain Dempster Shafer theory with a suitable example. **10**

5[a] What is Expert System? Explain its various components. Briefly describe 'MYCIN' Expert System. **10**

[b] Compare proposional logic and first order predicate logic. **4**

6[a] Show that the following sets of formulae are mutually consistent

(i0 $S_1 = \{ \sim PVQ, QVR, \sim (\sim P \wedge R)\}$

(ii) $S_2 = \{ P, P \wedge Q\}$ **4**

[b] What is resolution method? How it is used to find whether an argument is valid or not. **10**

7 Write short notes on any TWO

[a] Natural language processing

[b] Understanding

[c] Planning

[d] Semantic Net. **14**

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